

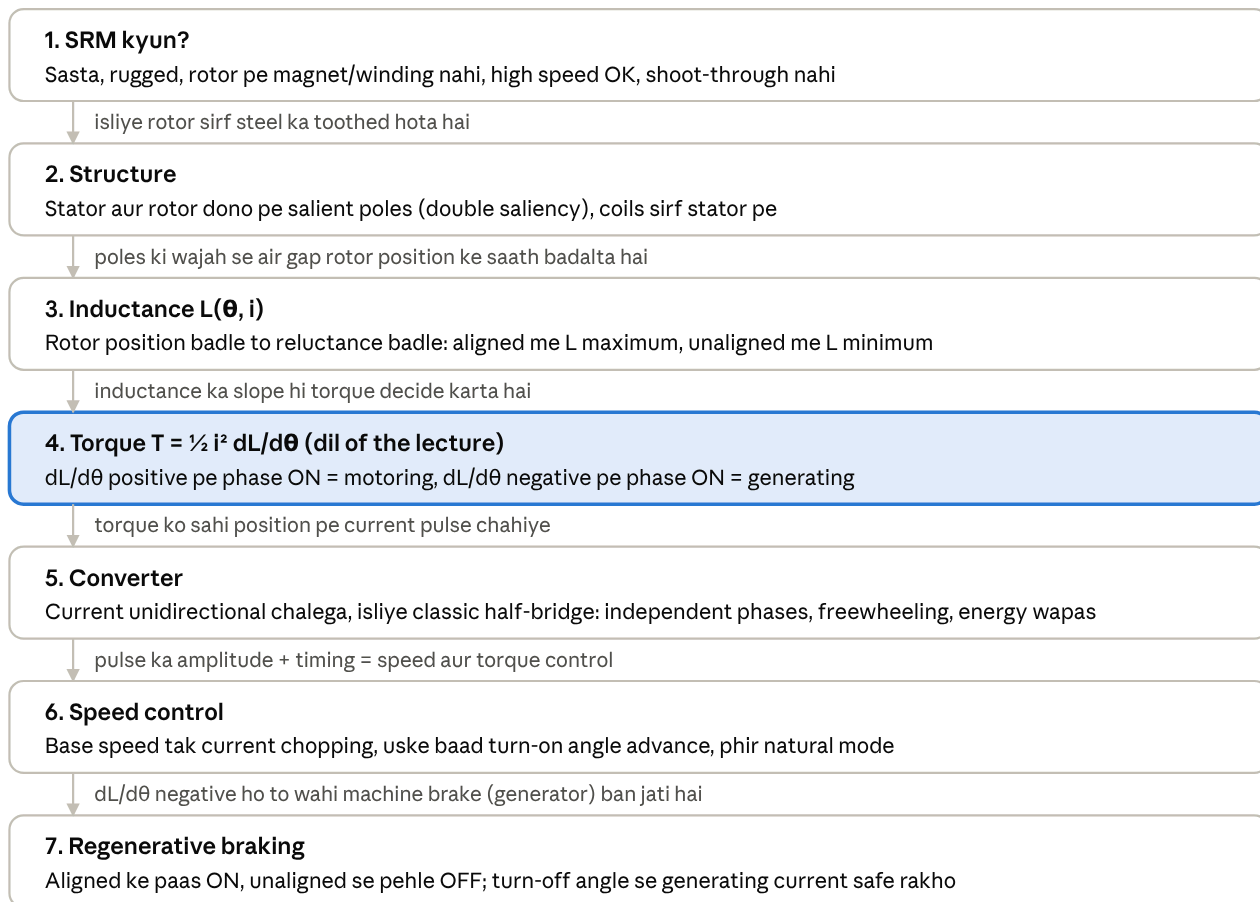
SRM Drive: Complete Hinglish Notes (Lecture 5, EEN5003)

Dr. Venu Sangwan, PEC. Beginner-friendly, saare pages ek story mein connected. Exam ke liye bilkul ready.

0. Poori Story ek nazar mein (Big Picture)

Pehle ye chain yaad kar lo. Poora lecture isi chain ko explain karta hai:

Poora lecture ek chain hai: har step agle step ki wajah banta hai



poori story · 7 steps, torque formula center mein

Upar se neeche padho: har box ka naya sawal agle box ka jawab banta hai, aur formula wala box (step 4) poore lecture ka center hai.

Ek line mein: SRM = rotor chaahta hai ki wo stator ke excited pole ke saath align ho jaye. Hum sahi time pe phase ON/OFF karke rotor ko lagatar "kheenchte" rehte hain. Bas yehi poora motor hai.

1. SRM Drive kyun attractive hai? (Page 1)

SRM drive **variable speed drives** ke liye achha candidate hai kyunki:

1. **Low cost**
2. **Rugged (mazboot) structure**
3. **Reliable converter topology**
4. **High efficiency over a wide speed range**
5. **Control simple hai**

Applications (exam mein zaroor likhna): EVs, HEV traction, aircraft starter/generator, mining drives, washing machines, door actuators.

Structure ki wajah se faayde

- Rotor pe **na PM (magnet) na winding**. Isse:
 - **Cost kam**
 - **High-speed operation possible.** Induction aur PM machines mein high speed pe **centrifugal force** se rotor ke magnets/windings ke toot ne ka darr hota hai. SRM mein ye problem nahi.

Inverter ki wajah se faayde

- Stator windings inverter ke **upper aur lower switch ke series mein** hoti hain (beech mein).
- Isse **shoot-through fault nahi hota.**
- **Shoot-through kya hai?** Ek hi leg ke upar aur neeche wale switch ek saath ON ho jayein to DC supply seedha short ho jata hai. Induction aur PM motor ke inverter mein ye risk hota hai. SRM mein winding dono switches ke beech hai, to dono ON hone pe bhi supply short nahi hoti, bas winding pe voltage aata hai.

Link: SRM kyun achha hai ye pata chal gaya. Ab ye poora drive ek system ki tarah dekho: motor ke saath converter, sensors aur controller kaise jude hain.

2. SRM Drive System: Block Diagram (Figure 1)

SRM drive = SRM + converter + sensors + controller, ek feedback loop me

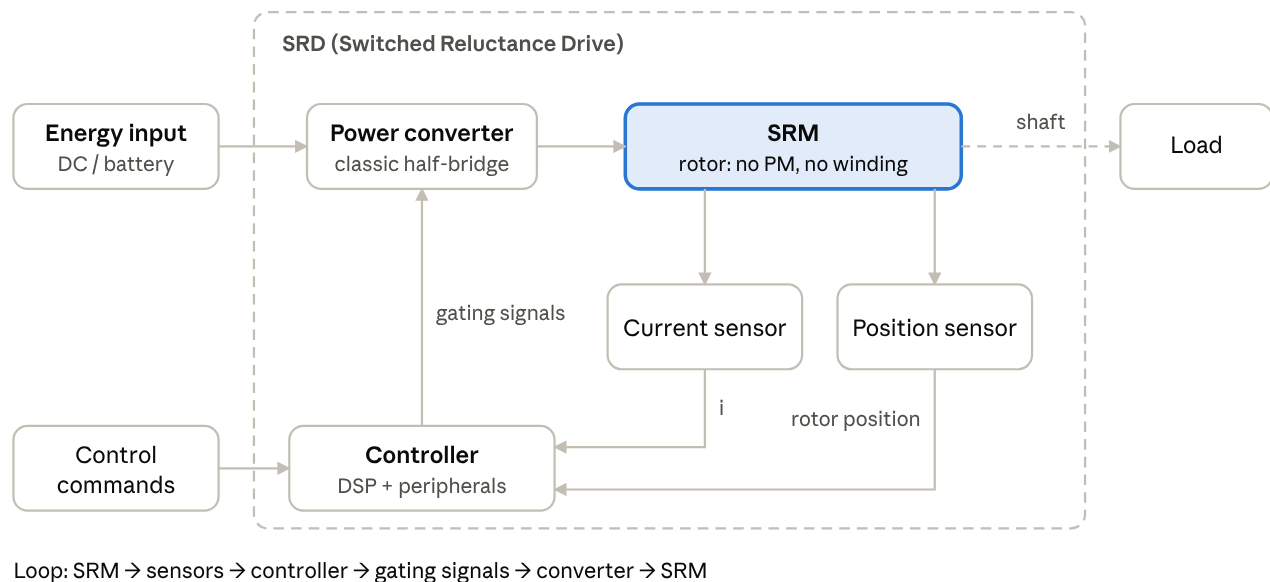


Figure 1 · SRM drive block diagram

Dashed dabba SRD hai. Sensors batate hain current aur rotor position, controller decide karta hai kaunsa switch kab ON ho, converter wahi karta hai.

Isko story ki tarah padho:

1. Battery/DC supply se **energy** aati hai.
2. **Power converter** us energy ko sahi phase ko deta hai.
3. **SRM** ghoomta hai aur **Load** ko chalata hai (dashed line = mechanical coupling).
4. SRM ke upar **Current sensor** aur **Position sensor** lage hain. Ye batate hain ki abhi current kitna hai aur rotor kahan hai.
5. **Controller** (DSP) in dono signals aur **control commands** (jaise speed set karo) ko dekhta hai.
6. Controller **gating signals** converter ko bhejta hai (kaunsa switch kab ON/OFF).
7. Ye **feedback loop** hai, isi se high performance milti hai.

Parts of conventional SRM drive: SRM, power inverter, sensors (voltage, current, position), control circuit (DSP controller + peripherals).

DC supply kahan se? Utility line se **front-end diode rectifier** ke through, ya **battery** se (EV mein battery).

Link: Block diagram me converter ke baad motor aata hai. Figure 2 isi connection ko zoom karke dikhata hai: phase windings aur inverter ke switches.

3. SRM aur uska Power Supply (Figure 2)

Figure 2 mein **8/6 SRM ke 4 phases (a, b, c, d)** dikhaye hain. Har phase ke do opposite stator poles hote hain (jaise a aur a'). Dono ki windings series mein judi hoti hain aur ek phase banati hain.

Inverter mein har phase ke liye:

- **Upar ka switch** (S_A1, S_B1, ...)
- **Neeche ka switch** (S_A2, S_B2, ...)
- **Do diodes** (D_A1, D_A2, ...) jo freewheeling ka kaam karte hain
- Winding in dono switches ke **beech** mein hai

Yehi **classic (asymmetric half-bridge) converter** hai, iska detail Section 8 mein.

Link: Windings stator pe hain, to ab motor ki body andar se dekho: kitne poles, aur rotor pe kya (aur kya nahi) hai.

4. Basic Magnetic Structure (Figure 3)

SRM ki body

- **Stator pe salient poles** (bahar nikle hue poles) aur **rotor pe bhi salient poles**. Isko **double saliency** kehte hain.
- **Stator pe concentrated windings** (har pole pe coil).
- **Rotor pe koi winding ya PM nahi**. Rotor sirf steel ka toothed gear jaisa hota hai.

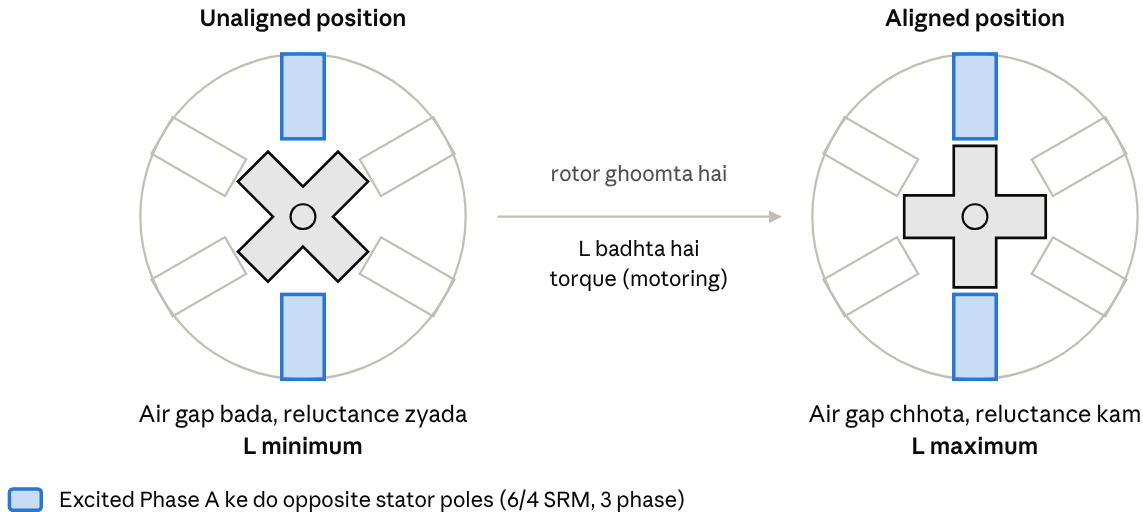
Common configurations

Configuration	Stator poles	Rotor poles	Phases
6/4 SRM (Fig 3a)	6	4	3

Configuration	Stator poles	Rotor poles	Phases
8/6 SRM (Fig 3b)	8	6	4

(Phases = stator poles / 2, kyunki opposite poles ek phase banate hain.)

Rotor excited pole ke saamne aate hi inductance badhta hai



6/4 SRM cross section · unaligned vs aligned

Dono taraf teeth hain (double saliency). Excited stator pole ke saamne rotor tooth aaye to air gap kam, isliye inductance max. 8/6 SRM me bhi yahi idea hai, bas 8 stator aur 6 rotor poles.

Double saliency ka matlab (bahut important)

Stator aur rotor dono ke poles bahar nikle hone se **flux path ki reluctance rotor position ke saath badalti hai**.

Reluctance = magnetic circuit ka resistance. Jab rotor pole stator pole ke saamne aata hai to air gap chhota hota hai, reluctance kam, inductance zyada.

Link: Double saliency ka seedha result: rotor ghoomte to air gap badalta hai, isliye inductance badalta hai. Next section isi inductance ko detail me samjhata hai.

5. Inductance, Flux Linkage aur Saturation (Figures 4, 5, 6)

Do cheezein rotor/current ke saath badalti hain

1. **Rotor position (theta) ke saath:** double saliency ki wajah se.
2. **Phase current ke saath:** kyunki SRM ko **high saturation** pe design kiya jata hai.

Isliye **flux linkage, bulk inductance, incremental inductance**, teeno **position aur current** dono pe depend karte hain.

Do extreme positions

- **Aligned position (theta = 0 degree):** rotor pole bilkul stator pole ke saamne. Air gap minimum, **inductance maximum**.
- **Unaligned position (theta = 30 degree in 8/6 SRM):** rotor pole do stator poles ke beech ke gap mein. Air gap maximum, **inductance minimum**.

Rotor unaligned se aligned ki taraf jaye (fixed current pe)

- Air gap kam, reluctance kam
- **Inductance badhta hai**
- **Flux linkage badhta hai**

Current badhao (fixed rotor position pe)

- Flux path **saturate** hota jata hai, reluctance **kam** hoti hai (text ke hisaab se)
- **Bulk inductance girta hai** (Figure 5: 2A wali curve sabse upar, 6A sabse neeche)
- Lekin **flux linkage phir bhi badhta hai** kyunki excitation zyada hai (Figure 4)

Figure 4 (Flux linkage vs current)

Har rotor angle ke liye ek curve. Aligned (0 degree) wali curve sabse upar, unaligned (-30 degree) wali sabse neeche. Chhote current pe curves seedhi (linear), bade current pe jhuk jati hain (saturation).

Figure 5 (Bulk inductance vs rotor position)

Ghanti (bell) jaisi shape. Beech mein (aligned, 0 degree) peak, dono taraf (unaligned, plus/minus 30 degree) minimum. Current 2A, 4A, 6A badhne pe peak neeche aata hai.

Figure 6 (Incremental inductance)

Unsaturated ho to **incremental inductance = bulk inductance**. Saturation mein dono alag. 6A pe curve almost flat aur bahut chhoti (heavy saturation).

Definitions (exam mein aate hain)

- **Bulk inductance:** $L = \text{flux linkage} / \text{current}$
- **Incremental inductance:** $l = d(\lambda)/d(i) = L + i \, dL/di$

Unsaturated case: flux current ke saath linear, to $l = L$.

Link: L ka position aur current ke saath badalna ab voltage equation me daalte hain. Wahin se back EMF nikalta hai.

6. Voltage Equations

General phase voltage equation (m phases)

$$V_j = R i_j + \frac{d}{dt} \sum_{k=1}^m \lambda_{jk}, \quad \lambda_{jk} = L_{jk}(i_k, \theta) i_k$$

- V_j = phase j pe lagaya voltage
- i_j = phase j ka current
- R = per phase resistance
- λ_{jk} = phase j ka flux linkage jo phase k ke current se bana
- $\lambda_{jk} = L_{jk}(i_k, \theta) \times i_k$

Mutual inductance (phases ke beech) bulk inductance ke compare mein bahut chhota hota hai, isliye **ignore** kar dete hain.

Chain rule se expand karo

$$V_j = R i_j + \sum_{k=1}^m \left(\left(L_{jk} + i_k \frac{dL_{jk}}{di_k} \right) \frac{di_k}{dt} + i_k \frac{\partial L_{jk}}{\partial \theta} \omega \right)$$

Jab sirf ek phase ON ho

$$V = R i + \underbrace{\left(L + i \frac{dL}{di} \right)}_{\text{incremental inductance}} \frac{di}{dt} + \underbrace{i \frac{\partial L}{\partial \theta}}_{\text{back EMF}} \omega$$

Samajhne ka tareeka: Supply voltage teen kaam karta hai:

1. Resistance mein drop ($R i$)
2. Current badhane mein (inductance wala term)
3. **Back EMF** ($i \times dL/d\theta \times \omega$) jo speed (ω) ke saath badhta hai

Back EMF ka yaad rakhne wala point: back EMF = $i (dL/d\theta) \omega$. **Speed badhegi to back EMF badhega.** Ye aage base speed ke concept mein kaam aayega.

Link: Voltage equation me $i dL/d\theta$ wala term back EMF tha. Wahi $dL/d\theta$ ab torque ka formula banata hai.

7. Torque Production (Figures 7, 8)

Basic idea

Rotor hamesha koshish karta hai ki excited stator pole ke saath align ho jaye (minimum reluctance). Isi tendency se torque banta hai.

Energy aur Coenergy (Figure 7)

Flux (λ) vs Current (i) ke graph pe:

- Curve ke **upar wala area** = **Stored field energy (W_f)**
- Curve ke **neeche wala area** = **Coenergy (W_f')**

$$W_{f'} = \int_0^i \lambda \, di$$

Torque formula (coenergy se)

$$T = \left[\frac{\partial W_{f'}}{\partial \theta} \right]_{i=\text{constant}}$$

Unsaturated case (linear flux)

Curve seedhi line, **coenergy = stored energy**, aur:

$$T = \frac{1}{2} i^2 \frac{dL(\theta)}{d\theta}$$

(L = unsaturated bulk inductance)

Saturated case

Simple formula nahi chalega, integral use karna padega:

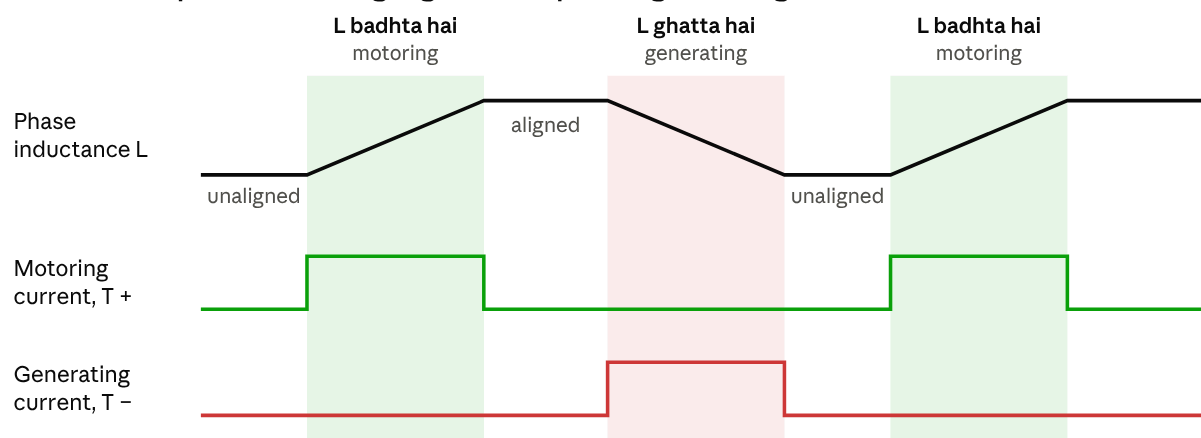
$$T = \int_0^i \frac{\partial L(\theta, i)}{\partial \theta} di$$

Is formula se 3 important conclusions

1. **Motoring (positive) torque tab milta hai jab $dL/d\theta > 0$** , matlab phase tab ON karo jab **inductance badh raha ho** (rotor pole stator pole ke paas aa raha ho).
2. T ka i^2 pe depend hai, isliye **current ki direction se fark nahi padta**. Matlab **current unidirectional** chalega.
3. Unidirectional current ki wajah se **sasta aur reliable inverter** use ho sakta hai (isi liye conventional AC bridge inverter nahi chahiye).

Figure 8: Ideal waveforms

L badhte waqt ON = motoring, L ghatte waqt ON = generating



$T = \frac{1}{2} i^2 \frac{dL}{d\theta}$: current ON hone ke time pe L ka slope torque ki sign decide karta hai.
Isliye position sensor zaroori hai, aur isliye unidirectional current kaafi hai.

Figure 8 · ideal inductance, current and torque

Hara zone = motoring window, laal zone = generating window. Generating mode motoring ka mirror image hai: wahi pulse, bas L ke girte hue slope pe.

- **L badh raha hai + current ON => positive (motoring) torque**
- **L ghat raha hai + current ON => negative torque (generating/braking)**
- **Isse pata chalta hai ki rotor position ka info zaroori hai** (isiliye Position sensor Figure 1 mein hai). Bina position jaane sahi time pe switch ON nahi kar sakte.

Total torque aur mechanical equation

```
\begin{aligned}
T_m &= \sum_{n=1}^N T(i, \theta) \\
T_m - T_l &= J \frac{d\omega}{dt} + B \omega \\
\omega &= \frac{d\theta}{dt}
\end{aligned}
```

- J = moment of inertia
- B = viscous friction
- Tl = load torque

Matlab: motor torque load se zyada ho to speed badhegi ($J d(\omega)/dt$), baaki friction ($B \omega$) mein jaata hai.

Link: Torque ke liye current pulse sahi position pe ON/OFF chahiye. Ye kaam converter karta hai, isliye ab converter topologies dekhte hain.

8. SRM Drive Converter (Figures 9, 10)

Kyun special converter?

Figure 8 se pata chala ki torque ko control karne ke liye **current pulse ka amplitude aur timing** rotor position ke saath synchronize karna padta hai. SRM mein current **unidirectional** hota hai, isliye **AC motors wala conventional bridge inverter use nahi hota**.

Figure 9: Teen-char main topologies

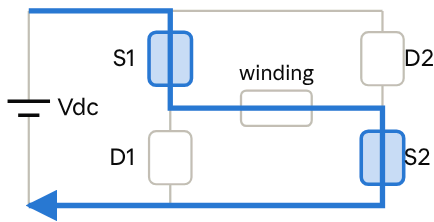
Fig	Name	Switches / Diodes	Main point
(a)	Classic half-bridge (asymmetric bridge)	2n switches + 2n diodes (n phases)	Sabse popular. Har phase independent control , high speed ke liye must
(b)	R-dump	1 switch + 1 diode per phase	Inefficient . Turn-off pe energy resistor R mein waste. Zero voltage mode nahi hota
(c)	(n+1) switch (Miller)	Saari phases ek switch share karti hain	Overlap between phases possible nahi , jabki high speed pe overlap zaroori hai
(d)	1.5n switch	Non-adjacent phases ke pairs switch share karte hain	(n+1) ki overlap problem solve. Sirf even number of phases ke liye
(e)	C-dump	Kam switches, ek capacitor C + switch Sc	Independent phase control milta hai. Disadvantage : turn-off pe phase coil pe negative voltage limited (V_c minus supply)

C-dump kaise kaam karta hai

Turn-off ke time stored magnetic energy **capacitor C** ko charge karti hai. Jab capacitor ka voltage ek value (V_c) tak pahunch jaye to energy **switch Sc** ke through wapas supply ko transfer ho jati hai.

Classic converter ke 3 modes (Figure 10) - Phase-1 example

Mode (a): Turn-on mode

(a) Turn-on mode: S1 aur S2 dono ON

Poora +Vdc winding pe lagta hai

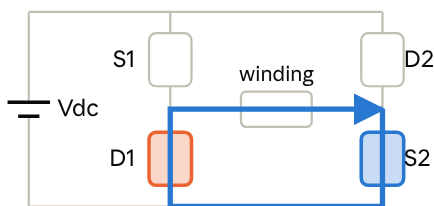
Current BADHTA hai

Path: +Vdc, S1, winding, S2, -Vdc

Supply se energy leti hai

Figure 10a · turn-on mode current path

- **S1 aur S2 dono ON.** Poora +Vdc winding pe lagta hai.
- **Current badhta hai.**

Mode (b): Zero voltage mode (freewheeling)**(b) Zero voltage mode: S1 OFF, S2 ON**

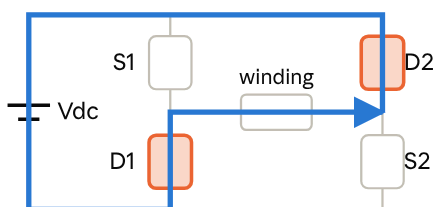
Current S2 aur D1 se freewheel karta hai

Winding pe voltage = 0

Na supply se energy leti, na wapas deti

Figure 10b · zero voltage (freewheeling) mode

- **S1 band, S2 ON.** Current S2 aur D1 se freewheel karta hai.
- **Winding na supply se energy leti hai na wapas deti hai.** Voltage = 0.

Mode (c): Turn-off mode**(c) Turn-off mode: S1 aur S2 dono OFF**

Current D2 aur D1 se supply ko jata hai

Winding pe negative bus voltage (-Vdc)

Current TEZI se girta hai

Fansi hui magnetic energy DC link ko wapas

Figure 10c · turn-off mode, energy back to the supply

Key: moti line = current ka rasta, rangeen dabba = ON switch ya conducting diode. Teeno modes mein S1 aur S2 ka on/off combination hi current regulate karta hai.

- **Dono switch OFF.** Current diodes D2 aur D1 se supply ke through behta hai.
- Winding pe **negative DC bus voltage (-Vdc)** lagta hai.
- **Magnetic circuit mein fansi energy DC link ko wapas jaati hai.**
- **Current tezi se girta hai.**

S1 aur S2 ko on-off karke phase current regulate hota hai.

Classic converter ke Advantages (exam point)

- Har phase **independent** control
- High speed pe jab adjacent phases ke currents **overlap** karte hain, tab bhi kaam karta hai
- Zero voltage mode available (efficiency achhi)
- Energy wapas supply ko jaati hai

Disadvantage: zyada switches ($2n$) => cost zyada.

Link: Converter current ko regulate kar sakta hai. Ab dekho speed badhne pe back EMF ke saath ye control kaise badalta hai (3 regions).

9. Modes of Operation (Figures 11, 12, 13)

Sabse pehle: Base Speed

Base speed = wo speed jahan back EMF = DC bus voltage.

Teen regions (torque-speed curve, Figure 13)

Torque-speed: constant torque, phir constant power, phir natural mode

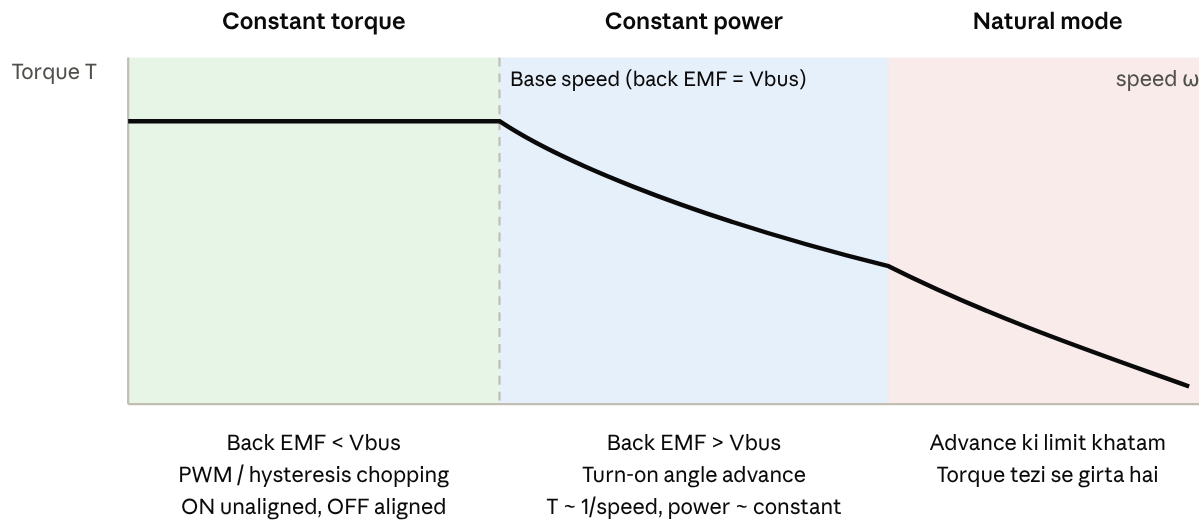


Figure 13 · torque-speed characteristic of the SRM

Base speed pe back EMF supply voltage ke barabar ho jata hai. Usse pehle current chopping se torque constant, uske baad turn-on angle advance karke power constant, aur advance ki limit ke baad natural mode.

Region 1: Below base speed => Constant Torque (Figure 11)

- Back EMF **bus voltage se kam** hai.
- Converter switches ON/OFF karke current **0 se rated value tak** regulate kar sakte hain.
- **Maximum torque tab milta hai jab phase unaligned position pe ON ho aur aligned position pe OFF ho**, aur current **rated value pe hysteresis ya PWM control se hold** ho.
- Waveform: phase voltage mein **+Vbus aur -Vbus ke chopping pulses**, current I_{max} aur I_{min} ke beech zigzag.

Region 2: Above base speed => Constant Power (Figure 12)

- Back EMF **bus voltage se zyada** ho jata hai.
- Jis position pe inductance ka slope positive hai, wahan **switch ON hone ke baad bhi current gir sakta hai** kyunki back EMF use limit karta hai.

- **Solution: turn-on angle ko advance karo.** Matlab phase ko **unaligned position se pehle hi ON kar do**, taaki current ko build hone ka zyada time mile. Jaise speed badhe, turn-on angle aur aage khisakta jata hai.
- Phir bhi speed badhne se back EMF badhta hai, current girta hai, **torque girta hai**.
- Is region mein **torque approximately 1/speed ke hisaab se girta hai** (reciprocal of speed), aur **maximum power approximately constant** rehta hai.
- Waveform: sirf **ek baar +Vbus**, phir **-Vbus**. Chopping nahi hoti (kyunki back EMF already current ko limit kar raha hai).

Region 3: Natural mode

- Turn-on ko advance karne ki bhi **limit** hoti hai: sirf wahan tak jahan tak **inductance ka slope negative** hai.
- Speed aur badhe to **koi phase advancing nahi bacha**, current build nahi kar sakte, **torque bahut tezi se girta hai**.
- Ise **natural mode** kehte hain.

Summary table

Region	Speed	Back EMF vs Vbus	Control	Torque	Power
Constant torque	Below base	Kam	Current chopping (PWM/hysteresis)	Constant (max)	Badhta
Constant power	Above base	Zyada	Turn-on angle advance	$\sim 1/\text{speed}$	$\sim \text{Constant}$
Natural	Bahut high	Bahut zyada	Koi advance nahi bacha	Tezi se girta hai	Girta hai

Link: Torque ka sign $dL/d\theta$ se decide hota hai. Slope negative pe ON karo to wahi motor brake lagakar generator ban jata hai: regenerative braking.

10. Generating Mode / Regenerative Braking (Figures 14, 15, 16)

Basic principle

Rotor hamesha minimum reluctance ki taraf jaana chahta hai.

- **Rotor pole excited phase ke paas aa raha hai** (inductance badh raha, positive slope): torque rotor ki direction mein => **Motoring mode**.
- **Rotor pole excited phase se door ja raha hai** (inductance ghat raha, negative slope): stator use roke rakhna chahta hai, torque rotor ki movement ke **ulta** => **Generating mode**.

EV mein kyun important?

Regenerative braking EVs aur HEVs ki propulsion drive ka important issue hai. Brake lagate waqt motor generator ban kar energy battery ko wapas deta hai.

Duality (motoring aur generating ka mirror)

Generating mode ke current waveforms motoring ke waveforms ka **mirror image** hote hain, **alignment position** ke around.

- **Motoring:** phase ON karo jab L badh raha ho (aligned position se pehle)
- **Generating:** phase ON aligned ke paas, OFF unaligned region se pehle

SRG (Switched Reluctance Generator) **singly excited machine** hai, isliye power lene ke liye usse **aligned position ke paas excite** karo aur **unaligned region se pehle turn off** karo.

Control in generating mode (Figure 14)

- **Low speed (a):** current level, turn-on aur turn-off angles teeno se control.
- **High speed, base se upar (b):** sirf turn-on aur turn-off angles se control.

SRG ka driving circuit (Figure 15)

Motor wale circuit jaisa hi, bas saath mein **capacitor C** aur **load Z_L** hai.

Kaise kaam karta hai:

- **Switches ON:** phase ko **supply** aur **capacitor** se energy milti hai.
- **Turn-off period:** motor ka freewheeling current **capacitor ko charge** karta hai aur **load ko energy** deta hai.
- **Start-up problem:** rotor pe **PM nahi** hai, isliye starting mein **external source (battery)** chahiye phase ko excite karne ke liye. Thodi der baad (transient ke baad) capacitor output voltage tak charge ho jata hai.
- External source ko charge hone ke liye design kar sakte hain, ya system operating point pe pahunche to **disconnect** bhi kar sakte hain.

Generating mode ki equations

```
\begin{aligned}
V_c - e &= L \frac{di}{dt} + R i, \quad e > 0 \quad ; ; (\text{phase-on}) \\
-V_c - e &= L \frac{di}{dt} + R i, \quad e < 0 \quad ; ; (\text{phase-off})
\end{aligned}
```

- V_c = inverter ka bus voltage (bus capacitor ka voltage)
- e = back EMF

Meaning: Generating region mein **back EMF negative** hota hai, isliye:

- Phase ON mein back EMF phase ko **bahut tezi se charge** karne mein madad karta hai.
- Turn-off mein back EMF **negative supply voltage ka opposition** karta hai, isliye current **dheere dheere** girta hai.

Danger: High speed + high load

Agar back EMF bus voltage se **zyada** ho jaye to **phase OFF** karne ke baad bhi current **badhta rehta hai**. Nuksan:

1. **Uncontrollable torque**
2. **Oversized converter** chahiye => cost aur size badhta hai
3. Prime mover ki speed vary karti hai, converter ko **worst case** ke liye design karna padta hai => aur zyada cost

Solution: Turn-off angle sahi chuno. Isse maximum generating current **safe region** mein aa jata hai.

Figure 16: Turn-off angle ka effect (8/6 SRM)

Teen curves: **Turn-off = 12, 13, 16 degree.**

- Turn-off = 12: peak current ~ **20 A**
- Turn-off = 13: peak current ~ **25 A**
- Turn-off = 16: peak current ~ **45-47 A**

Conclusion: Turn-off angle badhane se maximum generating current **bahut tezi se badhta hai**. Isliye safe operation ke liye turn-off angle chhota rakho.

11. EXAM CHEAT SHEET

Sabse important formulas

#	Formula	Meaning
1	$V_j = R i_j + d/dt (\sum \lambda_{jk})$	Phase voltage equation
2	$\lambda_{jk} = L_{jk}(i_k, \theta) i_k$	Flux linkage
3	$V = R i + (L + i dL/di) di/dt + i (dL/d\theta) \omega$	Single phase ON
4	Back EMF = $i (dL/d\theta) \omega$	Third term
5	$l = d(\lambda)/di = L + i dL/di$	Incremental inductance
6	$W_f' = \text{integral of } \lambda di$	Coenergy
7	$T = dW_f'/d(\theta) \text{ at constant } i$	Torque (general)
8	$T = (1/2) i^2 dL/d(\theta)$	Torque (unsaturated)
9	$T = \text{integral of } (dL(\theta, i)/d(\theta)) i di$	Torque (saturated)
10	$T_m = \sum T(i, \theta)$	Total torque
11	$T_m - T_l = J d(\omega)/dt + B \omega$	Mechanical equation
12	$\omega = d(\theta)/dt$	Speed-position
13	$V_c - e = L di/dt + R i \text{ (on); } -V_c - e = L di/dt + R i \text{ (off)}$	Generating mode

Possible exam questions aur 1-line answers

- SRM ke advantages?** Low cost, rugged, no PM/winding on rotor, high-speed capability, no shoot-through, high efficiency over wide speed, simple control.
- Rotor pe winding nahi, to torque kaise?** Reluctance torque: rotor excited stator pole ke saath align hone ki koshish karta hai.
- Motoring torque kab?** Jab phase tab ON ho jab inductance badh raha ho ($dL/d\theta > 0$).
- Generating torque kab?** Jab phase tab ON ho jab inductance ghat raha ho ($dL/d\theta < 0$).

5. **Position sensor kyun?** Kyunki torque ki sign phase ON karne ke position pe depend karti hai.
6. **Current unidirectional kyun chalega?** Torque i^2 pe depend hai.
7. **Classic converter ke fayde?** Independent phase control, overlap handle karta hai, zero voltage mode, energy recovery.
8. **Base speed kya hai?** Wo speed jahan back EMF = DC bus voltage.
9. **Above base speed kya karte hain?** Turn-on angle advance karte hain.
10. **Natural mode kya hai?** Jab advance karne ki limit khatam ho jaye aur torque tezi se gire.
11. **Torque-speed curve ke 3 regions?** Constant torque, constant power, natural.
12. **SRG ko battery kyun chahiye?** Rotor pe PM nahi hai, start-up mein excitation chahiye.
13. **Generating mode ka danger?** High speed/load pe back EMF > bus voltage, current control se bahar.
14. **R-dump kyun inefficient?** Energy resistor mein waste hoti hai, zero voltage mode nahi.
15. **C-dump ka disadvantage?** Negative voltage across phase limited (V_c minus supply).

Notes ki chhoti galtiyen (agar confuse ho to dhyan do)

- Page "Classic converter" mein "Figure 8a/8c" aur "Figure 6.10b" likha hai. Asal mein ye Figure 10 (a), (b), (c) hain.
- Figure 12 ka caption "Low-speed" hai, par waveform **high-speed (above base speed)** ka hai (single pulse, advanced turn-on).
- "Figure 6.69" ka matlab Figure 16 hai.
- **(n+1) switch converter** Figure 9c hai, aur **1.5n switch** Figure 9d hai (text mein sharing wala fix 9d bola gaya hai).
- **Incremental inductance formula** notes mein typo hai: sahi form $L = d(\lambda)/di$.

Ek line mein poora lecture

SRM mein rotor pe kuch nahi, dono taraf salient poles hain $\Rightarrow$ inductance rotor position ke saath badalta hai $\Rightarrow$ torque $= \frac{1}{2} i^2 \frac{dL}{d\theta}$ $\Rightarrow$ L badhte waqt phase ON karo to motoring, ghatte waqt ON karo to generating $\Rightarrow$ iske liye unidirectional current wala converter (classic half-bridge best) + position sensor + controller $\Rightarrow$ low speed pe current chopping, high speed pe turn-on angle advance, bahut high speed pe natural mode.

All the best for tomorrow! 🚀